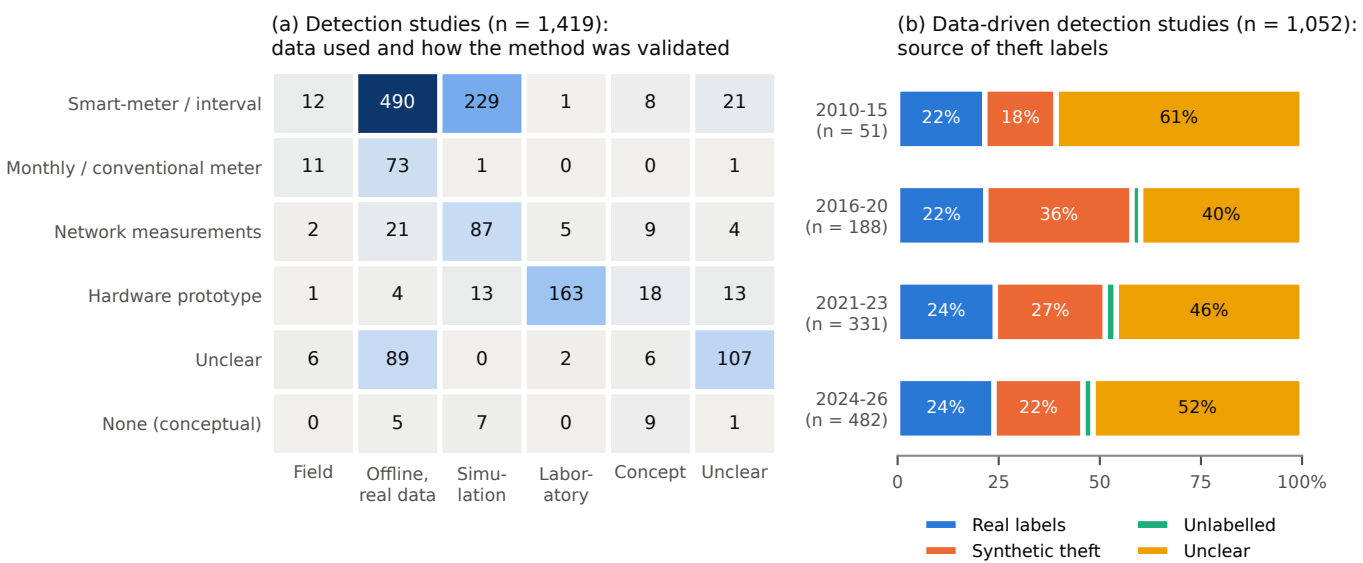
(a) Detection studies (n = 1,419):
data used and how the method was validated
Smart-meter / interval
Monthly / conventional meter
Network measurements
Hardware prototype
Unclear
None (conceptual)
12
490
229
1
8
21
11
73
1
0
0
1
2
21
87
5
9
4
1
4
13
163
18
13
6
89
0
2
6
107
0
5
7
0
9
1
Field
Offline, real data
Simu-lation
Labor-atory
Concept
Unclear
(b) Data-driven detection studies (n = 1,052):
source of theft labels
2010-15 (n = 51)
22%
18%
61%
2016-20 (n = 188)
22%
36%
40%
2021-23 (n = 331)
24%
27%
46%
2024-26 (n = 482)
24%
22%
52%
0
25
50
75
100%
Real labels
Synthetic theft
Unlabelled
Unclear